LAIEN ZOU

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EDUCATION

University of Toronto **Toronto, ON**
Honours Bachelor of Science (HBSc) Sep. 2025 – Expected Dec. 2028
Specialist in **Computer Science**; Specialist in **Data Science**

- **CGPA: 4.00/4.00** | Dean's List (2026)
- **Coursework:** Linear Algebra, Multivariable Calculus, Probability, Theory of Computation
- **Upcoming (Winter 2027):** Machine Learning (CSC311), Statistical Inference (STA261), Data Science I (JSC270)

PROJECTS

TTC Subway Delay Prediction | Python, pandas, scikit-learn | *GitHub* Oct. 2026 – Present

- Cleaned **45,000+** TTC subway delay records from Toronto Open Data, consolidating **1,251** inconsistent station names into 86 categories and excluding a target-leaking feature (Min Gap, r = 0.83 with delay length).
- Built logistic regression and random forest classifiers for long delays ($\geq$ 5 min) with a time-based split; tuning the decision threshold on a validation period raised F1 from 0.38 to **0.48**, outperforming class weighting (0.44) with half the false alarms.
- Found that infrastructure failures account for **55%** of long delays at 5–6 am vs. 14% at other times, and identified a 2025–2026 shift in delay rates, with F1 dropping from 0.50 on validation to 0.48 on later test data.

Steam Games Recommender (CSC111 Team Project) | Python, Graph Algorithms | *GitHub* Mar. 2026 – Apr. 2026

- Implemented the backend recommendation algorithm for **4,000+** Steam games, ranking similar games by **Jaccard similarity** over genres, tags, and platforms.
- Built review-quality filters (positive-review ratio and review count) to screen out poorly rated games before ranking.

Electricity Access in African Countries (STA130 Course Project) | R, Linear Regression Feb. 2026 – Mar. 2026

- Analyzed data from **54 countries** to study how electrification relates to poverty, population density, HDI, electricity cost, and per-capita consumption.
- Found a negative association between electrification and poverty rates, and lower access in low-density countries; used these findings to propose indicators for the Clean Air Task Force (CATF) electricity sector scorecard.

AWARDS

- **Later Life Learning Scholarship**, Innis College, University of Toronto Oct. 2026
- **Dean's List**, Faculty of Arts & Science, University of Toronto Jun. 2026
- **AP Scholar with Distinction**, College Board Aug. 2025

TECHNICAL SKILLS

- **Languages:** Python, Java, R, LaTeX
- **Libraries:** NumPy, pandas, scikit-learn, Matplotlib
- **Developer Tools:** Git, VS Code, PyCharm, IntelliJ, JupyterHub, RStudio
- **Spoken Languages:** English (Fluent), Mandarin (Native)